

The MOF-driven synthesis of supported palladium clusters with catalytic activity for carbene-mediated chemistry

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The development of catalysts able to assist industrially important chemical processes is a topic of high importance. In view of the catalytic capabilities of small metal clusters, research efforts are being focused on the synthesis of novel catalysts bearing such active sites. Here we report a heterogeneous catalyst consisting of Pd₄ clusters with mixed-valence 0/+1 oxidation states, stabilized and homogeneously organized within the walls of a metal-organic framework (MOF). The resulting solid catalyst outperforms state-of-the-art metal catalysts in carbene-mediated reactions of diazoacetates, with high yields (>90%) and turnover numbers (up to 100,000). In addition, the MOF-supported Pd₄ clusters retain their catalytic activity in repeated batch and flow reactions (>20 cycles). Our findings demonstrate how this synthetic approach may now instruct the future design of heterogeneous catalysts with advantageous reaction capabilities for other important processes.

Ultrasmall metal clusters^{1–5} have attracted a great deal of interest as extremely active and selective catalysts for organic reactions^{6,7}. Despite the efforts made, there is yet more work to be done to fully understand the exact nature of the catalytic species, their reactivity, structure, shape and nuclearity^{8,9}. Ligand stabilization¹⁰ and matrix-supported^{11,12} are the most efficient strategies to synthesize and characterize metal clusters. While the former allows structure determination of the cluster metal complex by single-crystal X-ray crystallography, these compounds rarely find applications in catalysis¹³ since ligand exchange with the reactants triggers decomposition of the cluster. In clear contrast, clusters supported on solids are stable catalytic species with highly reactive uncoordinated atoms¹, but only nuclearity and oxidation state of the metal by X-ray absorption spectroscopies combined with high-resolution microscopies can be addressed. Thus, the synthesis of supported metal clusters with precise topological and electronic properties, as occurs in metal complexes, would be of much interest.

In this context, metal-organic frameworks (MOFs)¹⁴ have emerged¹⁵ as very versatile materials for encapsulating catalytic nanosized species within their channels¹⁶. In addition, the regular and well-defined channels in MOFs appear, theoretically, as the perfect hosts to obtain information about the nature of the metallic catalyst species by means of single-crystal X-ray crystallography. However, after a number of studies on this topic showing small 1–2 nm metal nanoparticles^{17–20}, it is difficult to find reports of metal clusters below the nanometre size within MOF matrices.

Metal carbenoids are powerful catalytic and synthetic intermediates in organic synthesis, and different metals have been reported to

form and transfer carbenes²¹. Despite Pd compounds being found as catalysts for a plethora of fundamental transformations in organic synthesis, such as hydrogenations, carbon–carbon couplings, and many others²², they have been only occasionally employed as catalysts in carbene-transfer processes^{23–27}.

Here we report the preparation of Pd₄^{0/+1} clusters stabilized by the anionic framework of a MOF. The material is able to catalyse representative carbene-mediated reactions such as the inter- and intramolecular Buchner ring expansion reaction, the alcohol insertion and the dimerization of diazo compounds, to give some unique naturally occurring and synthetically useful products^{28–30}. This new catalytic behaviour of Pd emerges due to the stabilization of intermediate Pd⁺ valence within the supported-MOF cluster, which enables carbene bonding to give a catalytic activity that exceeds by one order of magnitude that of the state-of-the-art catalysts for these reactions. In addition, the solid catalyst can be recovered and reused up to 20 times in batch without any reactivation treatment or depletion of the catalytic activity, or operated in flow with solvent recycling during hours.

Synthesis and characterization of Pd₄-MOF

Herein, we take advantage of the emerging post-synthetic methods^{31,32} to synthesize (see Methods), in three steps, [Pd₄]²⁺ clusters from a preformed highly porous anionic MOF of formula Mg₂^{II}{Mg₄^{II}[Cu₂^{II}(Me₃mpba)₂]₃} · 45H₂O (**1**) [Me₃mpba^{4–} = N,N'-2,4,6-trimethyl-1,3-phenylenebis(oxamate)] (Fig. 1a) by: the already reported transmetalation³² of **1** to yield the more robust MOF Ni₂^{II}{Ni₄^{II}[Cu₂^{II}(Me₃mpba)₂]₃} · 54H₂O (**2**) (Fig. 1b); the

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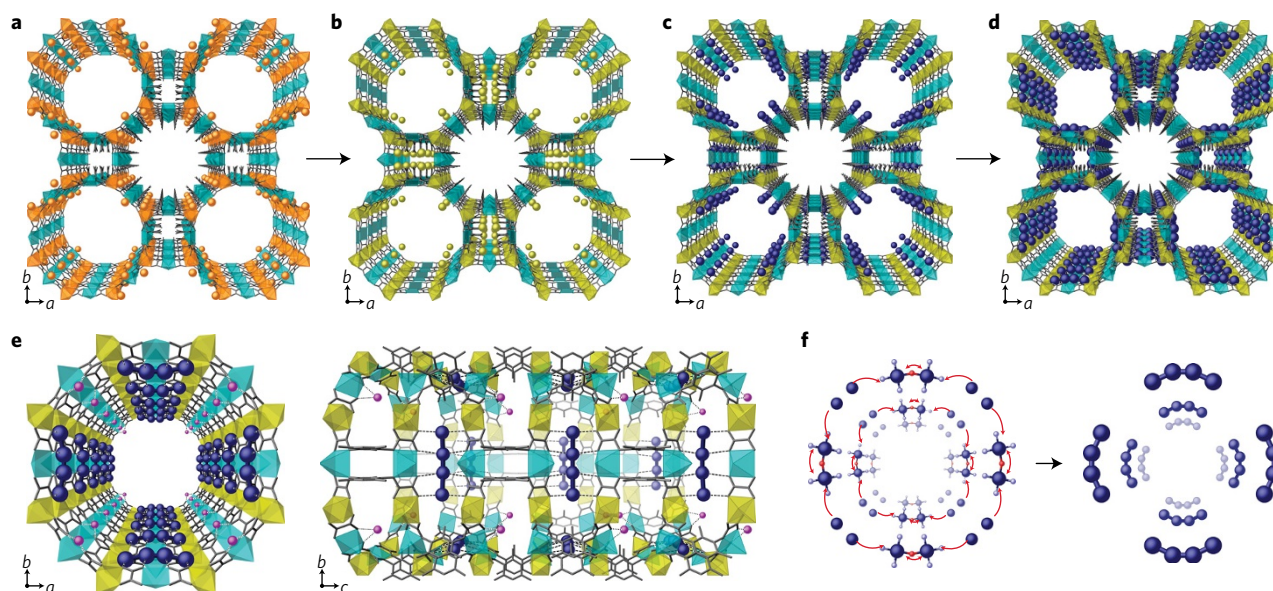


Figure 1 | X-ray crystal structure. **a–d**, Design approach showing the structures of **1–4** determined by single-crystal X-ray diffraction from the three-step post-synthetic process consisting of a transmetalation **1** (**a**) to give **2** (**b**), an exchange of the Ni^{II} cations of the pores by $[\text{Pd}^{\text{II}}(\text{NH}_3)_4]^{2+}$ ones yielding **3** (**c**) and the final reduction process affording **4** (**d**). **e**, Perspective views of a channel of **4** along the *c* (left) and *a* (right) axis. **f**, Detail of the proposed self-assembly after reduction process of the Pd^{II} units in **3** (left) to give the Pd_4 cluster in **4** (right). Copper, magnesium and nickel atoms from the network are represented by cyan, orange and yellow polyhedral, respectively, whereas organic ligands are depicted as sticks. Orange, yellow, pink, dark blue, pale blue and red spheres represent Mg, Ni, Na and Pd atoms and NH_3 and O^{2-} molecules/ions, respectively. Dashed and dotted lines represent the $\text{Pd} \cdots \text{O}$ and $\text{Na} \cdots \text{O}$ weak supramolecular interactions, respectively.

exchange of the nickel(II) cations hosted in the pores of **1** by $[\text{Pd}^{\text{II}}(\text{NH}_3)_4]^{2+}$ ones yielding the novel compound $[\text{Pd}^{\text{II}}(\text{NH}_3)_4][\text{Pd}_2^{\text{II}}(\mu\text{-O})(\text{NH}_3)_6(\text{NH}_4)_2]_{0.5}[\text{Ni}_4^{\text{II}}[\text{Cu}_2^{\text{II}}(\text{Me}_3\text{mpba})_2]_3] \cdot 52\text{H}_2\text{O}$ (**3**) (Fig. 1c); and the reduction process to give the final compound $[\text{Pd}_4]_{0.5}@\text{Na}_3[\text{Ni}_4^{\text{II}}[\text{Cu}_2^{\text{II}}(\text{Me}_3\text{mpba})_2]_3] \cdot 56\text{H}_2\text{O}$ (**4**) (Fig. 1d).

Overall, the presence of Pd^{II} cations in a controlled stoichiometry^{33,34} together with the confined state provided by the anionic porous network seems to play a key role in gaining control of the size and mixed-valence nature of metal particles and thus, such small clusters as $[\text{Pd}_4]^{2+}$ units were formed. The robustness of the three-dimensional network allowed the resolution of the crystal structure of every single material to unambiguously visualize the single-crystal to single-crystal guest inclusion and the final observation of quasi-linear $[\text{Pd}_4]^{2+}$ clusters (Figs 1 and 2 and Supplementary Figs 1–6), reminiscent of ligand-stabilized palladium sandwich complexes reported before^{10,35,36}, which are undoubtedly stabilized by the network (Figs 1e and 2 and structural details section in the Supplementary Information).

The nature of the Pd_4 –MOF compound **4** was established by the combination of a variety of techniques including: density functional theory calculations, inductively coupled plasma–mass spectrometry, elemental, powder X-ray diffraction (PXRD) and thermo-gravimetric analyses, Fourier transform infrared (FTIR) under CO and X-ray photoelectron (XPS) spectroscopies and scanning (SEM) and high-resolution transmission electron (HR-TEM) microscopies, including high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) measurements (Fig. 3, Supplementary Figs 7–17 and Supplementary Table 1). The N_2 adsorption isotherm at 77 K confirmed the permanent porosity of **4** (Supplementary Fig. 18). Finally, the resolution of the crystal structure of **4** allowed an unambiguous observation of such small Pd entities in **4**.

The anionic $\text{Ni}_4^{\text{II}}\text{Cu}_6^{\text{II}}$ open-framework structure displays hydrophilic octagonal pores, exhibiting a virtual diameter of 2.2 nm, that accommodate free nickel(II) cations that can be exchanged afterwards to give **3** and **4** (Figs 1b–d and 2, and Supplementary

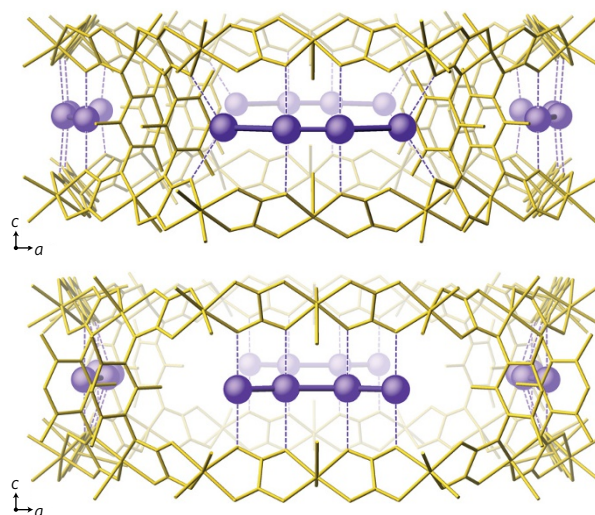


Figure 2 | Cluster structure. Perspective view along the *b* crystallographic axis of two portions (top and bottom) of the crystal structure of **4** enclosing the two crystallographic not equivalent ligand-stabilized quasi-linear $[\text{Pd}_4]^{2+}$ NCs. MOF structure is depicted as gold sticks and Pd atoms are represented as purple spheres. Purple dotted lines show some of the cluster-network interactions.

Figs 1, 2 and 4a) with no evidence of remarkable changes in structural parameters (Supplementary Table 2)³². The arrangement of Pd^{2+} moieties in **3** and that of Pd nanoclusters (NCs) in **4**, confined into the channels, are strictly related. In **3** they exhibit either mononuclear, $[\text{Pd}^{\text{II}}(\text{NH}_3)_4]^{2+}$, or dinuclear complexes of the type $[\text{Pd}_2^{\text{II}}(\mu\text{-O})(\text{NH}_3)_6]^{2+}$ (Supplementary Fig. 3), whereas in **4**, after reduction, they finally assemble in centrosymmetric tetranuclear Pd_4 NCs (Fig. 1 and Supplementary Figs 1b, 2 and 4). The Pd^{2+} ions from the oxo-bridged dinuclear entities in **3**

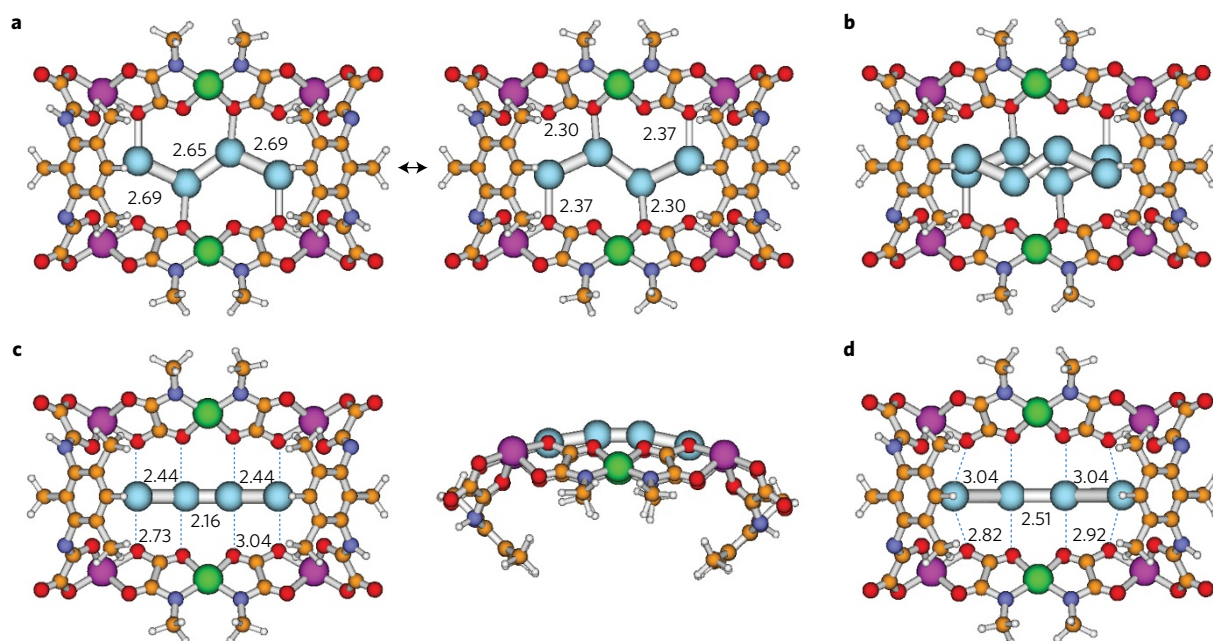


Figure 3 | Density functional theory calculations. Optimized geometry of Pd_4 in a $\text{Pd}_4\{\text{Ni}_4[\text{Cu}_2(\text{Me}_3\text{mpba})_2]_3\}^{2-}$ model. **a–d**, The two equivalent structures separated (**a**), overlapped (**b**) and averaged (**c**) are shown, together with the XRD structure (**d**). Pd–Pd and Pd–O (italics) distances are given in Å. Pd, Cu, Ni, N, O, C and H atoms are depicted in blue, green, pink, purple, red, orange and white, respectively.

(Supplementary Fig. 3) are tetracoordinated, in a distorted square planar geometry, with Pd–NH₃ and Pd–O bond distances similar to those found in the literature³⁷. The Pd atoms in **4** are two- or one-coordinated, respectively in a slightly bent geometry [Pd–Pd–Pd angles of 155(1)° and 169(1)°] (Supplementary Fig. 4) growing tetranuclear NCs. The inner Pd–Pd bond length values [2.57(3)/2.44(3) Å] are significantly shorter than the relatively long outer ones [3.16(2)/2.91(3) Å] although both falling in the range of Pd–Pd bonding interactions^{10,35,36,38}. This asymmetry of bond lengths within the $[\text{Pd}_4]^{2+}$ moiety is a striking feature, because despite the similarity in the shape of the Pd_4 cluster, such a difference was not observed in the two previously reported ligand-stabilized tetrapalladium compounds, where Pd–Pd separation varies in the range 2.538(1)–2.563(1) Å (ref. 35) and 2.571(1)–2.622(1) Å (ref. 36).

The possibility to have disordered $[\text{Pd}_2]$ instead of $[\text{Pd}_4]$ clusters might be consistent and it has been considered. However, diverse factors determine that Pd_4 clusters are, by far, the most probable resolution (see structural details section in the Supplementary Information). To support the nature of these clusters, theoretical calculations were carried out using a simplified model of a $\text{Pd}_4\{\text{Ni}_4[\text{Cu}_2(\text{Me}_3\text{mpba})_2]_3\}^{2-}$ unit (Fig. 3 and Supplementary Fig. 7). When Pd atoms are allowed to move without restrictions within the octagonal ring model, they approach the donor oxygen atoms of the wall forming bonds with typical Pd–O distances of 2.30/2.37 Å, and optimized Pd–Pd distances of 2.69 and 2.65 Å for the outer and inner bonds, respectively (Fig. 3a). Despite this result apparently contradicting the quasi-linear arrangement of the $[\text{Pd}_4]^{2+}$ moiety observed by XRD, averaging of the two equivalent structures obtained from optimization yields a slightly bent, Pd_4 unit with longer outer Pd–Pd distances (Fig. 3b,c), in agreement with experiment (Fig. 3d) where the positions of Pd atoms were refined with considerably large thermal motion. Analysis of charge distribution reveals a mixed-valence state of the type $\text{Pd}^+\text{Pd}^0\text{Pd}^0\text{Pd}^+$, with a calculated net atomic charge of 0.34 and 0.18 e on the outer and inner Pd atoms, respectively.

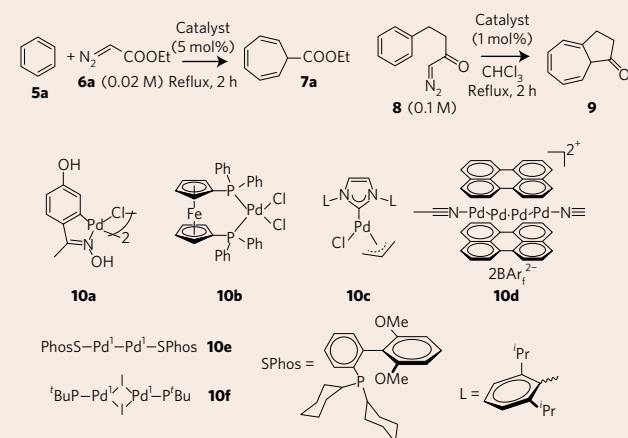
Figure 1f suggests a tentative formation mechanism of the Pd_4 NCs by proposing that the $[\text{Pd}_2^{\text{II}}(\mu\text{-O})(\text{NH}_3)_6]^{2+}$ complexes act as

precursors of the central NCs core, whereas the mononuclear Pd^{2+} cations become the terminal atoms in the Pd_4 NCs (Supplementary Fig. 3e). Finally, hydrated charge-counterbalancing alkali Na^+ cations are present in **4** as a result of the reduction with NaBH_4 . They are distributed within the channels, further stabilizing the whole net (Supplementary Fig. 5).

The Pd^{2+} -based guest molecules in **3** are efficiently organized within the MOF pore, where they can react, aggregating in ordered dimers. This suggests that the crystalline porous host aligns the molecules within its cavities through a high degree of molecular recognition. While in **3**, $[\text{Pd}_2^{\text{II}}(\mu\text{-O})(\text{NH}_3)_6]^{2+}$ dimers interact with the net through $-\text{NH}_3 \cdots \text{O}$ hydrogen bonds (Supplementary Fig. 6), in **4**, tetranuclear $[\text{Pd}_4]^{2+}$ clusters are most likely stabilized either by the network, through weak supramolecular interactions between Pd and oxamate moieties, or through a probable hydrated primary coordination sphere around Pd atoms, hydrogen-bonded to the net (Figs 1f and 2 and Supplementary Fig. 4). Despite the typical structural uncertainty related to the difficulty to determine the positions of solvent molecules in porous materials, the most plausible $[\text{Pd}_4]^{2+}$ cluster structure is solvated and stabilized by a synergic effect between the network and the solvent molecules surrounding clusters. The large accessible free voids (56% of the total cell volume) allow the guest to become more than an innocent bystander. Such a vastly solvated nano-confined space stabilizes catalytic NCs without protective groups while retaining access to reactant species.

PXRD patterns of **3** and **4** (Supplementary Fig. 8) confirm the crystallinity and pureness of the bulk samples with no trace of the typical peaks of Pd NPs, not even after catalysis.

FTIR spectroscopy of adsorbed CO (ref. 39) is able to differentiate between supported Pd^0 , Pd^+ and Pd^{2+} . The spectrum of **4** (Supplementary Fig. 10a) shows absorption bands at 2,140 cm^{-1} and 1,882 cm^{-1} , which correspond to linearly adsorbed CO to Pd^+ and Pd^0 , respectively, together with some weak bands assignable to Pd^{2+} metal cations^{39–41}. For the sake of comparison, the FTIR spectrum of adsorbed CO on the mono-directional zeolite LH after impregnation and reduction of Pd^{2+} was recorded (Supplementary Fig. 10b), showing that the mono-directional zeolite is able to stabilize Pd^+ to a much lesser extent.

Table 1 | Catalytic activity and structures of Pd complexes.

Entry	Catalyst	Yield 7a (%)	Yield 9 (%)
1	Pd(OAc) ₂	9	<1
2	PdCl ₂	15	<1
3	PdCl ₂ (NH ₃) ₄ · H ₂ O	<5	<1
4	Pd(OAc) ₂ phen	7	<1
5	PdCl ₂ (PhCN) ₂	<5	<1
6	PdCl ₂ (PPh ₃) ₂	11	<1
7	Pd ₂ dba ₃	17	4
8	Nájera's complex 10a	<5	<1
9	Ferrocene-Pd ²⁺ 10b	<5	<1
10	Pd-NHC complex 10c	<5	<1
11	Pd ⁰ black	9	<1
12	Pd ²⁺ -C (5 wt%)	<5	<1
13	Pd ⁰ -C (5 wt%)	<5	<1
14	Pd ⁰ -Al ₂ O ₃	<5	<1
15	Pd ²⁺ -MgO	<5	<1
16	Pd ⁰ -MgO	<5	<1
17	Pd ²⁺ -HY	<5	1
18	Pd ²⁺ -NaY	<5	<1
19	[Pd(NH ₃) ₄] ²⁺ -NaY	<5	<1
20	Pd ⁰ -NaY	<5	<1
21	[Pd(NH ₃) ₄] ²⁺ -ZSM5	<5	<1
22	Pd ⁰ -HZSM5	<5	<1
23	Pd ²⁺ -KL	14	3
24	Pd ⁰ -KL	34	4
25	Pd ²⁺ -Mordenite	18	1
26	Pd ⁰ -Mordenite	26	2
27	3	68	17
28	4	75	57
29	Perylene-Pd ₄ ²⁺ 10d	<5	<1
30	Barder's complex 10e	<5	<1
31	(^t Bu ₃ PPdI) ₂ 10f	<5	<1
32*	4 (0.5 mol%)	53 (91)	39 (71)
33	Rh ₂ (OAc) ₄ (0.5 mol%)	52 (71)	83 (99)

Reaction conditions: **6** (15 µl, 0.14 mmol, inter-) or **8** (100 mg, 0.60 mmol, intra-) in the corresponding solvent (6 ml) at reflux temperature with the Pd catalyst (5 and 1 mol%, respectively) for 2 h. Under the indicated reaction conditions, conversion of the diazoacetate is always 100%, either to the desired products, alkene dimers, protonation to the acetate, and/or unidentified decomposition products. Reaction yields were triple-checked by *in situ* ¹H-NMR analysis in deuterated solvents after calibration with the pure products, GC-MS analysis after derivatization to the more stable conjugated isomers with CF₃COOH or Et₃N and calibration with the pure products⁴⁹, and isolated yields. The given yields refer to the average with a ±5% error. *6 h reaction time; pump addition of the diazocompound in brackets.

The XPS spectrum of compound **3** is shown in Supplementary Fig. 11a. The Pd3d line consists of only one doublet with binding energy (BE) of Pd3d_{5/2} peak equal to 338.6 eV, corresponding to Pd(II) (ref. 42). In turn, the XPS of **4** (Supplementary Fig. 11b), after

deconvolution of the Pd3d line, shows the typical peak for Pd⁰ at a binding energy of BE(Pd3d_{5/2}) = 335.7 eV together with a non-ordinary signal at BE(Pd3d_{5/2}) = 337.0 eV, an energy between that of Pd⁰ and the residual signals of Pd²⁺ at BE(Pd3d_{5/2}) = 338.7 eV, and with a 1:1 ratio with respect to Pd⁰, which, even if distinguishing between Pd⁰ and Pd⁺ is challenging, can be attributed to Pd⁺, as previously reported⁴².

All these results, together, further support that both Pd⁺ and Pd⁰ can be found in **4** and that the Pd⁺Pd⁰Pd⁰Pd⁺ arrangement predicted by theoretical calculations is the most probable electronic state of the cluster.

Aberration-corrected HR-TEM images of **4** (Supplementary Figs 12a, 13 and 14) reveal no aggregation of the Pd₄ NCs and their uniform distribution throughout the highly crystalline framework. Analysis of these images also directly visualizes the lattice fringes of the net, where the interplanar spacing is 0.248 nm (Supplementary Fig. 12a), corresponding to (110) planes of the MOF, and confirms its retained crystallinity. Both SEM and HAADF-STEM imaging and their corresponding EDX elemental mappings for Cu, Ni, Pd and Na elements (Supplementary Figs 12b,d and 15–17) confirm a homogeneous spatial distribution of the Pd₄ NCs through the MOF crystal.

Pd₄-MOF catalyses carbene-transfer reactions

Table 1 shows the catalytic results of different Pd compounds for the Buchner reaction of benzene derivatives to give naturally occurring and synthetically useful cycloheptatrienes²⁸. This unique transformation is catalysed only by dirhodium salts²⁹ or particular copper complexes³⁰. Here, from the different Pd compounds tested that include salts, complexes, pure metallic forms, and different Pd supported species (entries 1–26), only **3** and **4** (entries 27–28) catalyse efficiently the intermolecular Buchner reaction of benzene **5a** and ethyl diazoacetate **6a** (Table 1 left) and the intramolecular reaction of **8** (Table 1 right).

Kinetic analysis showed that the green solid **3** gives an induction time of ~30 min after which the intermolecular reaction starts and proceeds (Supplementary Fig. 19), and *in situ* magic angle spinning–solid nuclear magnetic resonance (MAS-NMR) of a sample of **3** treated with an equimolecular amount respect to Pd of isotopically labelled **6a** N¹³CH–COOEt (refs 43,44), in gas phase, shows the immediate transformation of **6a** to the corresponding dimer (Supplementary Fig. 20) with the recovered solid active for the Buchner reaction without induction period. These results point to diazoacetate **6a** as a reducing agent of Pd²⁺ in **3**, to give **4** under reaction conditions, as NaBH₄ does externally. The solid precursors **1** and **2** do not show any catalytic activity, which confirms that the catalytic activity comes from [Pd₄]²⁺ and not from any defects on the MOF structure.

Zeolites able to stabilize Pd⁺ cluster species within their anionic zeolite framework, such as KL and mordenite, give significant yields of **7a** (entries 23–26), which supports the active role of Pd⁺ in the reaction. In contrast, ligand-stabilized complexes such as a perylene-stabilized linear [Pd₄]²⁺ complex (**10d**; refs 35,45) and two commercially available phosphine-stabilized Pd⁺ dimer complexes, **10e** (ref. 46) and **10f** (ref. 47), were prepared and evaluated catalytically (entries 29–31). In all cases, product **7a** was not observed, which can be easily understood taking into account that the labile perylene ligand (**10d**) is also a potential reactant in the Buchner reaction and that **10d–f** are temperature-sensitive compounds.

The catalytic amount of **4** could be lowered to 0.5 mol% and reused up to 20 times without a significant depletion of its inherent catalytic activity for the intermolecular reaction (entry 32, Supplementary Fig. 21). The process could also be run in flow (Supplementary Fig. 22) by adding **6a** with a syringe pump to a fixed amount of catalyst **4** under recirculating benzene **5a**. In this way,

Table 2 | Catalytic activity and substrate scope.

R ¹	R ²	R ³	Product	Yield (%)	R ¹	R ²	R ³	Product	Yield (%)
H	H	Et	7a	91	OMe	H	Et	7g	92
Me	H	Et	7b	57	OMe	H	^t Bu	7h	46
Me	Me	Et	7c	64	H	H	Bn	7i	73
F	H	Et	7d	78	Me	H	Bn	7j	70
CF ₃	H	Et	7e	48	OMe	H	Bn	7k	95
H	H	^t Bu	7f	32	F	H	Bn	7l	65

R ³	R ⁴	Catalyst (mol%)	Time (min)	Product	Yield (%)	E/Z
Me	Me	Pd ₂ dba ₃ (0.5)	10	12a	72	-
Me	-			13a	10	40/60
Me	Me	10f (0.5)		12a	59	-
Me	-			13a	15	47/53
Me	Me	Pd ⁰ -KL (0.5)		12a	53	-
Me	-			13a	21	52/48
Me	Me	Pd ⁰ -Mor (0.5)		12a	44	-
Me	-			13a	6	50/50
Me	Me	4 (0.5)		12a	98	-
		4 (0.005)	120		99	- (TON = 49,500)
		4 (0.001)			65	- (TON = 65,000)
Me	-	4 (0.5)	10	13a	95	50/50
		4 (0.005)	120		96	50/50
		4 (0.001)			93	50/50 (TON = 93,000)
Me	^t Amyl	4 (0.005)		12b	74	
^t Bu	Me			12c	88	
Bn	Me			12d	96	
Bn	^t Amyl			12e	85	
8	CD ₃ OD	4 (0.5)		12f	99	-
^t Bu	-	4 (0.005)		13b	78	50/50
Bn	-			13c	97	65/35
8	-	4 (0.05)		13d	98	50/50

Top, scope for the Pd₄-catalysed intermolecular Buchner reaction. Bottom, catalyst comparison and substrate scope for other Pd₄-catalysed carbene-mediated reactions of diazoacetates. Under the indicated reaction conditions, conversion of the diazoacetate is always 100%, either to the desired products, protonation to the acetate, and/or unidentified decomposition products.

a solution of ~1 g of **7a** in 100 ml of benzene (~0.1 M) could be obtained, which not only saves fivefold amounts of benzene but also enables a safer handling of **6a** for industrial applications^{30,48}. Besides, the recovered catalyst **4** completely retained its intrinsic catalytic activity after working in flow mode for 24 h (Supplementary Fig. 23). A turnover number (TON) > 2,000 was obtained for **4** either in batch or in flow, a value one order of magnitude higher than the state-of-the-art catalyst Rh₂(OAc)₄ (TON = 104, entry 33), which rapidly decomposes to inactive species after the first use²⁹.

Table 2 shows that **4** catalyses the intermolecular Buchner reaction of different arenes and diazoacetates in yields comparable to Rh₂(OAc)₄ (ref. 29). In contrast to the soluble catalyst, the isomeric distribution obtained with **4** for products **7a–l** shifts significantly towards less hindered products (Supplementary Table 3). For instance, the isomeric ratio at full conversion for the 2-, 3- and

4-methyl substituted products in **7b** changes from ~1:1:2.5 with Rh₂(OAc)₄ or ~1:1:6 with Rh₂(OAc_{3-F})₄ (ref. 29) to 0:1:1 with **4**, indicating that the MOF structure imposes shape selectivity effects during the catalytic process. Indeed, the intramolecular Buchner reaction of the bulkier diazoacetate **8** (ref. 49) proceeds in 57% yield, a significantly lower value than the intermolecular reaction, since the channel of **4** probably restricts sterically the transition state. Note that no conversion of **8** was found with KL and mordenite with smaller ~7 Å² channels.

The reaction rate for the intermolecular Buchner reaction of **5a** (Supplementary Fig. 24) follows a first-order kinetics with respect to both reactants and catalyst to give the equation rate $v_0 = k[\text{cat}][\text{5a}][\text{6a}]$ (k = observable kinetic constant) at low concentrations of **6a**. The participation of a Pd carbenoid in the rate-determining step of the reaction is supported by:

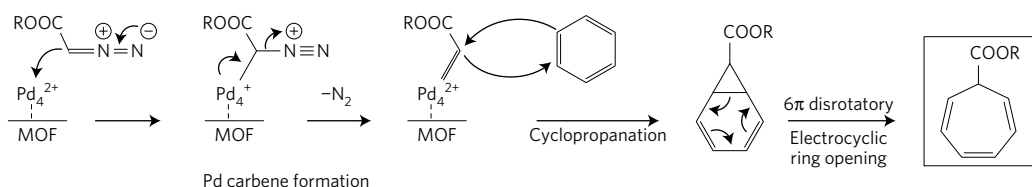


Figure 4 | Proposed mechanism for Pd₄-catalysed intermolecular Buchner reaction.

the similar rates found for N₂ liberation and formation of cycloheptatriene **7a**; values for $\Delta H^\ddagger$ and $\Delta S^\ddagger$ of 19.0(8) kcal mol⁻¹ and -9.9(5) kcal mol⁻¹ K⁻¹, respectively, obtained by an Eyring plot of the intermolecular reaction and which agree with previous reports on associative transition states during this reaction⁵⁰; and the structure of the conjugated cycloheptatrienes obtained here can only come, in principle, from the classically accepted cyclopropanation/electrocyclic ring opening route, since other potential ways of coupling/opening should lead to different products. Thus, a plausible mechanism for the Pd₄-catalysed Buchner reaction can be proposed (Fig. 4). The reaction would start with the formation of a Pd carbenoid, which enables the cyclopropanation of the arene and the 6 π disrotatory electrocyclic ring opening of the so-formed norcaradiene, to give the final cycloheptatriene product.

Finally, Table 2 also compares the catalytic activity of different Pd compounds for other representative carbene-transfer-mediated reactions of diazocompounds, such as the insertion of alcohols or the dimerization. The results show that 0.005 mol% of **4** gives >90% yield of products **12a-f** and **13a-d** with TONs approaching 100,000 catalytic cycles in some cases, thus outperforming not only other Pd catalysts but also the state-of-the-art metal catalysts for these reactions^{29,30}.

Conclusions

The confinement and protection provided by the MOF together with an exquisite control of the stoichiometry of the Pd²⁺ cations has allowed the multigram-scale synthesis of linear Pd₄^{0/1+} clusters, homogeneously distributed and stabilized along the MOF channels and fully characterized with X-ray crystallography. These ultrasmall clusters catalyse very efficiently representative carbene-mediated reactions such as the inter- and intramolecular Buchner reaction, the O-H insertion and the dimerization of diazocompounds, with TONs at least one order of magnitude higher than the state-of-the-art most active catalysts. The solid Pd₄^{0/1+}-MOF catalyst is very stable under reaction conditions, recoverable and reusable, and can work in a batch or in flow with no observable loss of activity with time. These results bring Pd to the selected group of metals that catalyse carbene-mediated reactions.

Methods

Methods, including statements of data availability and any associated accession codes and references, are available in the [online version of this paper](#).

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Author contributions

E.P., A.L.-P., A.C. and D.A. designed the research; E.P. and J.F.-S. coordinated the whole work; E.R.F.-P. and M.M. performed synthetic work; D.A. performed powder and single-crystal XRD characterization and analysed data; A.L.-P. carried out the catalytic experiments; J.G. and D.O. performed spectroscopic characterization and analysed data; J.M.H. performed microscopy measurements; M.B. carried out the theoretical calculations; E.P., A.L.-P., A.C., D.A., J.F.-S., M.B. and J.G. wrote and revised the paper.

Additional information

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Competing financial interests

The authors declare no competing financial interests.

Methods

Materials. All chemicals were of reagent-grade quality. They were purchased from commercial sources and used as received. Crystals of $\text{Mg}_2^{\text{II}}\{\text{Mg}_4^{\text{II}}[\text{Cu}_2^{\text{II}}(\text{Me}_3\text{mpba})_2]_3\} \cdot 45\text{H}_2\text{O}$ (1), $\text{Ni}_2^{\text{II}}\{\text{Ni}_4^{\text{II}}[\text{Cu}_2^{\text{II}}(\text{Me}_3\text{mpba})_2]_3\} \cdot 54\text{H}_2\text{O}$ (2) and $[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2$ were prepared as previously reported (see Supplementary Methods). Alternatively, a large-scale synthesis of $\text{Ni}_2^{\text{II}}\{\text{Ni}_4^{\text{II}}[\text{Cu}_2^{\text{II}}(\text{Me}_3\text{mpba})_2]_3\} \cdot 54\text{H}_2\text{O}$ (2) was carried out by direct reaction of two aqueous solutions (250 ml each) of $\text{Na}_4[\text{Cu}_2(\text{Me}_3\text{mpba})_2] \cdot 4\text{H}_2\text{O}$ (8.71 g, 0.010 mol) and $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (3.88 g, 0.013 mol) and subsequent addition, after filtration and re-suspension in 150 ml water of the resulting compound, of 1.94 g (0.0067 mol) of $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (yield 99%).

Syntheses. For $[\text{Pd}^{\text{II}}(\text{NH}_3)_4][\text{Pd}_2^{\text{II}}(\mu\text{-O})(\text{NH}_3)_6(\text{NH}_4)_2]_{0.5}\{\text{Ni}_4^{\text{II}}[\text{Cu}_2^{\text{II}}(\text{Me}_3\text{mpba})_2]_3\} \cdot 52\text{H}_2\text{O}$ (3), well-formed deep green prisms of 3, which were suitable for XRD, were obtained by immersing crystals of 2 (about 5 mg, 0.0015 mmol) for 48 h in 5 ml of a $[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2$ aqueous solution (0.004 mmol). Aiming at industrial applications, a multigram-scale procedure was also carried out by using the same synthetic procedure but with greater amounts of both, a powder sample of compound 2 (20 g, 5.8 mmol) and $[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2$ (3.43 g, 14.0 mmol), with the same successful results and a very high yield (20.33 g, 96%). Anal.: calcd (%) for $\text{Cu}_6\text{Ni}_4\text{Pd}_2\text{C}_{78}\text{H}_{189}\text{N}_{20}\text{O}_{88.5}$ (3,652.3): C, 25.65; H, 5.22; N, 7.67. Found: C, 25.68; H, 5.13; N, 7.33. IR (KBr): $\nu = 3,014, 2,951$ and $2,913\text{ cm}^{-1}$ (C–H), $1,607\text{ cm}^{-1}$ (C=O).

For $[\text{Pd}_4]_{0.5}@\text{Na}_3\{\text{Ni}_4^{\text{II}}[\text{Cu}_2^{\text{II}}(\text{Me}_3\text{mpba})_2]_3\} \cdot 56\text{H}_2\text{O}$ (4), both, crystals (about 5 mg) and a powder polycrystalline sample of 3 (about 10 g), were suspended in 50 ml of a $\text{H}_2\text{O}/\text{CH}_3\text{OH}$ (1:2) solution to which an excess of NaBH_4 , divided into 26 fractions (each fraction consisting of 1 mole of NaBH_4 per mole of 3 to give a final NaBH_4/MOF molar ratio of 26 or, which is the same, NaBH_4/Pd atom molar ratio of 13), was added progressively in the space of 72 h. After each addition, the mixture was allowed to react for 1.5 h. After this period, samples were gently washed with a $\text{H}_2\text{O}/\text{CH}_3\text{OH}$ solution and filtered on paper giving excellent yields (98%). Anal.: calcd (%) for $\text{Cu}_6\text{Ni}_4\text{Pd}_2\text{Na}_3\text{C}_{78}\text{H}_{172}\text{N}_{12}\text{O}_{92}$ (3,648.1): C, 25.68; H, 4.75; N, 4.61. Found: C, 25.68; H, 4.63; N, 4.65. IR (KBr): $\nu = 3,013, 2,964$ and $2,914\text{ cm}^{-1}$ (C–H), $1,603\text{ cm}^{-1}$ (C=O).

Single-crystal X-ray diffraction. Crystal data for 3 and 4: tetragonal, space group $P4/mmm$, $T = 90(2)\text{ K}$, $Z = 4$; 3: $\text{C}_{78}\text{H}_{189}\text{Cu}_6\text{N}_{20}\text{Ni}_4\text{O}_{88.5}\text{Pd}_2$, $a = 35.920(2)\text{ Å}$, $c = 15.3561(9)\text{ Å}$, $V = 19,813(2)\text{ Å}^3$, $\rho_{\text{calc}} = 1.224\text{ g cm}^{-3}$, $\mu = 1.259\text{ mm}^{-1}$; 4: $\text{C}_{78}\text{H}_{172}\text{Cu}_6\text{N}_{12}\text{Na}_3\text{Ni}_4\text{O}_{92}\text{Pd}_2$, $a = 35.794(13)\text{ Å}$, $c = 15.063(5)\text{ Å}$, $V = 19,298(16)\text{ Å}^3$, $\rho_{\text{calc}} = 1.256\text{ g cm}^{-3}$, $\mu = 1.299\text{ mm}^{-1}$. Further details can be found in the Supplementary Information.

X-ray powder diffraction. PXRD patterns of solid polycrystalline samples 3 and 4, before and after catalysis, were obtained with a Empyrean PANalytical powder diffractometer, using $\text{Cu K}\alpha$ radiation ($\lambda = 1.54056\text{ Å}$) at room temperature ($2\theta = 2^\circ - 60^\circ$).

FTIR spectroscopy of adsorbed CO. The spectra were recorded on a Biorad FTS-40A spectrometer equipped with a DTGS detector. The experiments have

been carried out in a home-made IR cell able to work in the high- and low- (77 K) temperature range. Further experimental details can be found in the Supplementary Methods.

Microscopy. High-resolution transmission electron microscopy (HR-TEM), high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) and energy-dispersive X-ray analysis (EDX) characterizations were done using a HAADF-FEI-TITAN G2 electron microscope. Five micrograms of the material was re-dispersed in 1 ml of absolute ethanol. Carbon-reinforced copper grids (200 mesh) were submerged into the suspension 30 times and then allowed to dry in air for 24 h.

Scanning electron microscopy coupled with energy-dispersive X-ray (SEM/EDX) was carried out with an XL 30 ESEM (PHILIPS) microscope equipped with a home-made EDX detector.

X-ray photoelectron spectroscopy. Samples were prepared by dropping a solid water suspension onto a molybdenum plate followed by air drying, and then measurements were performed on a SPECS spectrometer equipped with a Phoibos 150 MCD-9 analyser using a non-monochromatic Mg KR (1,253.6 eV) X-ray source working at 50 W. As an internal reference for the peak positions in the XPS spectra, the C1s peak has been set at 284.5 eV.

Magic angle spinning–solid-state NMR. Fifty micrograms of catalyst 3 was introduced into glass inserts and dehydrated at 80°C for 18 h. Then, $4.7\text{ }\mu\text{mol}$ of isotopically labelled **6a** $\text{N}_2^{13}\text{CH-COOEt}$, corresponding to $\text{Pd}/\textbf{6a} = 1$, was introduced onto the dehydrated material. The glass inserts were sealed while they were immersed in liquid nitrogen. *In situ* ^{13}C solid-state NMR spectra were recorded at room temperature with a Bruker AVIII HD 400 WB spectrometer. The glass inserts were fitted into 7 mm rotors and were spun at 5 kHz in a Bruker BL7 probe. ^{13}C CP/MAS NMR spectra were recorded with proton decoupling, with a ^1H 90° pulse length of 5 μs , and a recycle delay of 3 s.

Computational details. All calculations are based on density functional theory and were carried out with the Gaussian 09 program package, using the hybrid B3PW91 functional, the LANL2DZ basis set for Pd, Cu and Ni atoms, and the standard 6-311G(d,p) basis set by Pople for O, N, C and H atoms. Atomic charges were calculated using the natural bond order (NBO) approach. A detailed description of models and methods is given in the Supplementary Information.

Catalytic experiments. A detailed description of all the catalytic experiments is given in the Supplementary Information.

Data availability. All data supporting the findings of this work are available from the corresponding authors on request. Crystallographic data for the structure reported in this paper have been deposited in the Cambridge Structural Database with CCDC numbers 1517224–1517225.